

# Modulation and Modeling of Monoclonal Antibody N-Linked Glycosylation in Mammalian Cell Perfusion Reactors

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**ABSTRACT:** Mammalian cell perfusion cultures are gaining renewed interest as an alternative to traditional fed-batch processes for the production of therapeutic proteins, such as monoclonal antibodies (mAb). The steady state operation at high viable cell density allows the continuous delivery of antibody product with increased space-time yield and reduced in-process variability of critical product quality attributes (CQA). In particular, the production of a confined mAb N-linked glycosylation pattern has the potential to increase therapeutic efficacy and bioactivity. In this study, we show that accurate control of flow rates, media composition and cell density of a Chinese hamster ovary (CHO) cell perfusion bioreactor allowed the production of a constant glycosylation profile for over 20 days. Steady state was reached after an initial transition phase of 6 days required for the stabilization of extra- and intracellular processes. The possibility to modulate the glycosylation profile was further investigated in a Design of Experiment (DoE), at different viable cell density and media supplement concentrations. This strategy was implemented in a sequential screening approach, where various steady states were achieved sequentially during one culture. It was found that, whereas high ammonia levels reached at high viable cell densities (VCD) values inhibited the processing to complex glycan structures, the supplementation of either galactose, or manganese as well as their synergy significantly increased the proportion of complex forms. The obtained experimental data set was used to compare the reliability of a statistical response surface model (RSM) to a mechanistic model of N-linked glycosylation. The latter

outperformed the response surface predictions with respect to its capability and reliability in predicting the system behavior (i.e., glycosylation pattern) outside the experimental space covered by the DoE design used for the model parameter estimation. Therefore, we can conclude that the modulation of glycosylation in a sequential steady state approach in combination with mechanistic model represents an efficient and rational strategy to develop continuous processes with desired N-linked glycosylation patterns.

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**KEYWORDS:** N-linked glycosylation; mammalian cell culture; perfusion bioreactor; mathematical model; monoclonal antibody

## Introduction

Driven by the advantages connected to continuous steady-state processing, perfusion bioreactors are becoming increasingly popular for the production of recombinant therapeutic glycoproteins (Langer, 2014). When dealing with mammalian cell cultures, a shift from batch to continuous manufacturing can be beneficial for the entire process, including increased volumetric productivity, manufacturing flexibility, reduced equipment size, and most importantly product quality (Karst et al., 2017; Konstantinov and Cooney, 2015). Since fed-batch cultivations rely on concentrated but often unbalanced feeds to prevent nutrient depletion at high cell densities, cells are exposed to a constantly changing environment and accumulation of toxic byproducts. This, in combination with the long product residence time inside the bioreactor, favors undesired chemical and enzymatic degradation as well as modifications that can affect critical quality attributes (CQA) of the product (Gramer and Goochee, 1993). In contrast, the presence of an outlet flow in perfused systems allows removing toxic

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byproducts and results in high cell density cultures with short and adjustable product residence times. This results in a minimization of protein CQA heterogeneity combined with an increased space time yield (Godawat et al., 2015; Warikoo et al., 2012), although typically at the expenses of higher media consumption.

Among the CQAs, N-linked glycosylation has been proven to affect the therapeutic performance of monoclonal antibodies (mAb), including stability, immunogenicity, circulatory lifetime, and therapeutic efficacy in a direct structure/function relationship (del Val et al., 2010; Eon-Duval et al., 2012; Hossler, 2012). Available methods to control the glycosylation of recombinant proteins have been extensively reviewed (Brühlmann et al., 2015; Hossler et al., 2009; Shi and Goudar, 2014) and include host cell engineering (Butler and Spearman, 2014; Kanda et al., 2007; Malphettes et al., 2010; Umaña et al., 1999), process conditions (Ahn et al., 2008; Butler, 2006; St. Amand et al., 2014b, 2016) as well as medium components (Ivarsson et al., 2014). Most of these studies have been carried out in fed-batch mode, where the role of different glycosylation-modulating factors is inevitably affected by variations in the cellular environment throughout the entire cultivation (Villiger et al., 2016c). Examples include important C-sources such as glucose (Liu et al., 2014; Villiger et al., 2016c), accumulation of waste products such as ammonia (Gawlitsek et al., 2000; Yang and Butler, 2000), depletion of intracellular nucleotide sugar donors (Kochanowski et al., 2008; Wong et al., 2010), and possibly changes in gene expression (St. Amand et al., 2014a). In this respect, perfusion cultures represent an interesting alternative allowing to independently investigate the impact of individual parameters on the mAb N-linked glycosylation of mammalian cell cultures (Ahn et al., 2008). However, such studies are limited due to the extensive duration of the operations and the lack of high throughput perfusion systems.

In this work, we investigate the N-linked glycan distribution of a human recombinant monoclonal antibody produced in CHO cells using a continuous Alternating Tangential Flow (ATF)-equipped perfusion bioreactor. The potential of modulating the extent of different glycan forms is explored in a Design of Experiment (DoE) with varying viable cell densities (VCD) as well as the supplementation of manganese and galactose. The effect of these factors has been extensively documented in fed batches (Gramer et al., 2011; St. Amand et al., 2014a), but data in perfusion processes are limited. Variations of the VCD set point in perfusion are inevitably linked to changes in metabolite concentrations of the culture broth. In particular, waste product metabolites have been shown to affect glycosylation, with ammonia being one of the most effective inhibitors of glycosylation (Gawlitsek et al., 2000). Manganese and galactose support the production of more complex glycans displaying galactose or sialic acids moieties as terminal units.  $Mn^{2+}$  is a primary cofactor for N-acetylglucosaminyltransferase-I (GnTI), -II (GnTII), and  $\beta$ -galactosyltransferase (GalT) (Gramer et al., 2011; Ramakrishnan et al., 2004), while galactose is the sugar precursor of the UDP-Gal nucleotide-sugar donor. To develop a strategy that allowed to efficiently screen different process conditions for modulating glycosylation while reducing the experimental effort required, we combined sequential screening and DoE approaches.

The application of mathematical models for the development and optimization of continuous cell cultures is particularly useful, since small-scale high throughput technologies of perfusion cultures are still in

development. Given the mechanistic understanding of the biosynthetic pathway of N-linked glycosylation, the development of mechanistic mathematical models is feasible (Hossler et al., 2007; Jedrzejewski et al., 2014; Jimenez del Val et al., 2011; Krambeck and Betenbaugh, 2005; Umaña and Bailey, 1997). In this regard, a mechanistic model describing monoclonal antibody N-linked glycosylation as a function of process conditions in continuous cultures was adapted from fed-batch model developed by our group and compared to a statistical response surface model (RSM) (Villiger et al., 2016b). The predictive capabilities of both models were tested by comparison with additional experiments conducted within and outside the operating process parameter space used for tuning the model parameters.

## Experimental

### Perfusion Cell Culture

A proprietary CHO cell line (Merck Serono SA, Switzerland) producing a fully humanized monoclonal antibody and proprietary cell culture media (Merck Serono SA, Switzerland) were used for all culture experiments. After thawing, cells were expanded in a chemically defined media using shake tubes at 36.5°C, 320 rpm, and 90% humidity for 1 week in a shaking incubator (Adolf Kühner AG, Switzerland). A perfusion seed bioreactor (Infors AG, Switzerland, 2 L working volume) was employed to increase cell densities to  $40 \times 10^6$  cells/mL in 1 week. Subsequently, cells were transferred to two identical perfusion bioreactors (DASGIP<sup>®</sup>, Eppendorf AG, Switzerland) each equipped with an ATF 2 device (Repligen Corporation, Waltham, MA) (Karst et al., 2016). A detailed description of the reactor dimensions can be found elsewhere (Karst et al., 2016). Polyether sulfone (PES) hollow fiber modules (Spectrum Laboratories Inc., Rancho Dominguez, CA) (area: 1570 cm<sup>2</sup>, fiber diameter: 1 mm, pore size 0.5  $\mu$ m) were mounted on the ATF diaphragm pump, which was operated at an effective flow rate of 1.5 L/min. Cells were cultured at pH 7.1, 50% dissolved oxygen tension (DOT) and 36.5°C in a 1.5 L scale. The DOT set point was maintained by adapting the oxygen fraction supplied through an open pipe sparger at a gas flow rate of 0.22 vvm. Sufficient mixing was ensured by stirring the culture broth at 400 rpm with a Rushton turbine impeller. The cell density was controlled by automated manipulation of the bleed rate. In all culture experiments, the harvest rate was fixed at a constant volume exchange rate. The fresh media feed was adjusted according to a gravimetric feedback to keep the reactor weight constant. Depending on VCD set point a fixed fraction of a base and enriched media were set to ensure the nutrient supply (Karst et al., 2017). Antifoam reagent (Gibco FoamAway<sup>™</sup>, Life Technologies) was added on demand to prevent foam built up.

### Constant Long-Term Culture

Before starting to investigate the effect of various parameters on the glycosylation of the produced mAb, long term experiments at  $40 \times 10^6$  cells/mL, referred to as constant run 1 and 2 in the following, were performed for about 26 days. There were two objectives regarding the constant long-term cultures: First, the proof of maintaining stable operation for about a month and second, to investigate how much time the systems needs to achieve

steady state, both with respect to the cell environment as well as product quality. Inoculation at the VCD set point was achieved using the seed bioreactor. While the bleed rate was controlled automatically according to the online VCD measurement, the harvest rate was maintained constant at 1.33 reactor volumes per day (RV/Day). A mixture of 65% base, 35% enriched feed media, and 46.6 mmol/L glucose was used in the fresh media feed.

Modulation of Glycosylation—Experimental Design

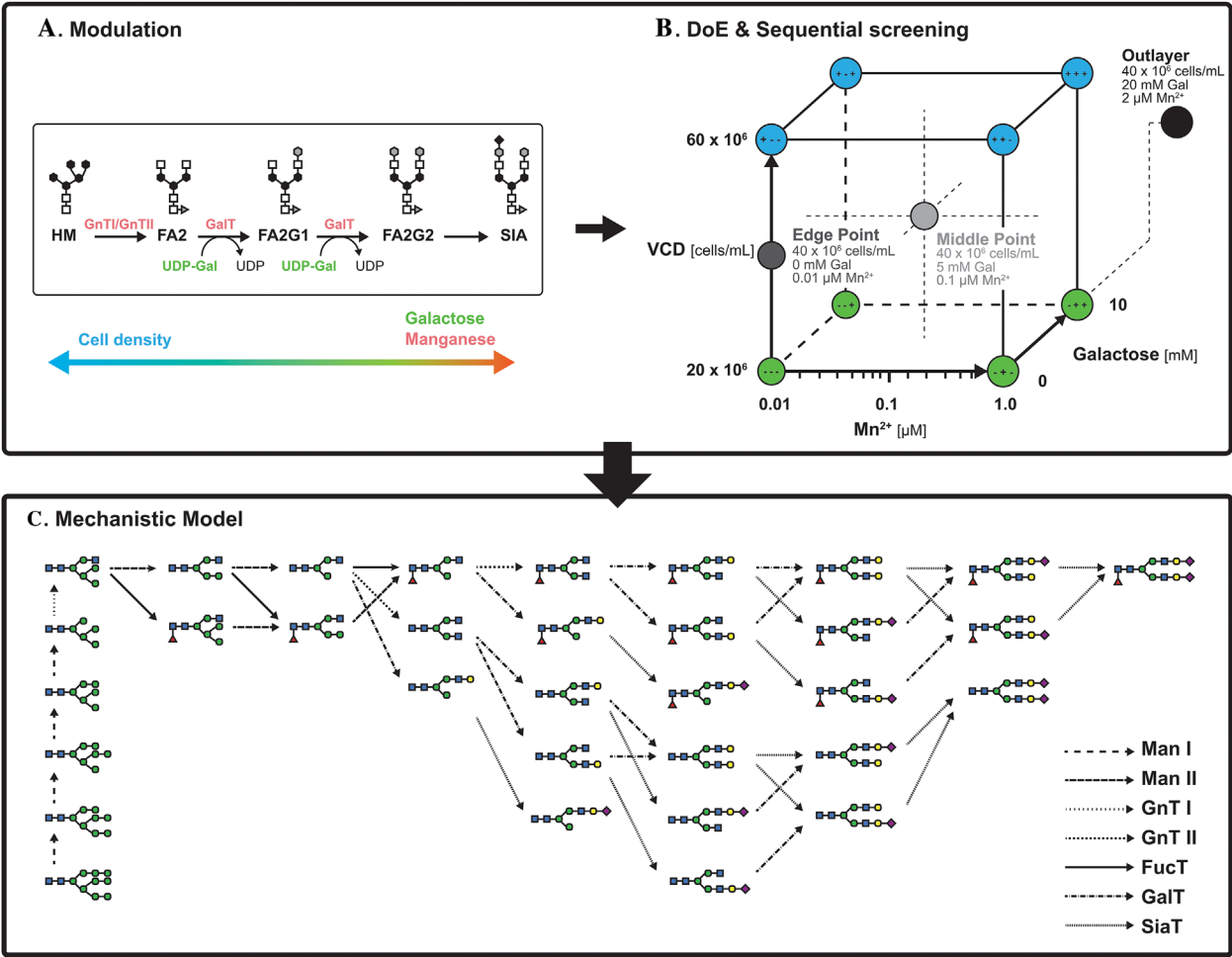
A DoE based on a two levels—full factorial design with varying viable cell density (VCD), MnCl<sub>2</sub>, and D-Galactose levels was performed in sequential perfusion culture experiments as shown in Figure 1. The single experimental conditions are summarized in Table I. All perfusion cultures were set up and operated as described in section Perfusion cell culture, with a fixed harvest rate of 1 RV/day. Low (20 × 10<sup>6</sup> cells/mL) and high (60 × 10<sup>6</sup> cells/mL) VCD were set using the biocapacitance probe, whereas the fraction of base to enriched media was set to 100/0% and 30/70%, respectively to account

for the higher metabolite consumption. MnCl<sub>2</sub> (Sigma–Aldrich, Switzerland) and D-Galactose (Thermo Fisher Scientific, Steinhausen, Switzerland) were added to the fresh media to achieve the final concentrations listed in Table I. The first eight conditions of Table I were used to fit a RSM and a mechanistic model of N-linked glycosylation for perfusion culture (details provided in section Mathematical Model). The potential of describing operating points in and outside of the full factorial design was investigated at the last conditions listed in Table I. These conditions included the middle point (40 × 10<sup>6</sup> cells/mL, 0.1 μM MnCl<sub>2</sub>, 5 mmol/L Galactose), an outlier (40 × 10<sup>6</sup> cells/mL, 2 μmol/L MnCl<sub>2</sub>, 20 mmol/L Galactose), and an edge point (40 × 10<sup>6</sup> cells/mL, 0.01 μmol/L MnCl<sub>2</sub>, 0 mmol/L Galactose).

Analytics

Offline Measurements

Cell density and viability were measured by trypan blue exclusion (Cedex, Roche Diagnostics, Germany). Glucose and lactate



**Figure 1.** Schematic representation of the paper line-out. (A) Modulating impact of galactose, manganese, and cell density on the mAb N-linked glycosylation machinery. (B) Full factorial Design of Experiments to investigate the effect of VCD, manganese, and galactose supplementation on N-linked glycan distribution. The experiments have been carried out according to a sequential screening of operating points. The obtained experimental data was fitted to both a mechanistic model and a response surface model. (2) Reaction scheme used to describe the N-glycosylation for the mAb in the mechanistic model.

**Table I.** Experimental conditions of the full factorial DoE varying viable cell density (VCD), galactose (Gal), and manganese ( $Mn^{2+}$ ).

| DoE condition | VCD (cells/mL)  | Gal (mmol/L) | $Mn^{2+}$ ( $\mu$ mol/L) |
|---------------|-----------------|--------------|--------------------------|
| — — —         | $20 \cdot 10^6$ | 0            | 0.01                     |
| — + —         | $20 \cdot 10^6$ | 10           | 0.01                     |
| — — +         | $20 \cdot 10^6$ | 0            | 1                        |
| — + +         | $20 \cdot 10^6$ | 10           | 1                        |
| + — —         | $60 \cdot 10^6$ | 0            | 0.01                     |
| + + —         | $60 \cdot 10^6$ | 10           | 0.01                     |
| + — +         | $60 \cdot 10^6$ | 0            | 1                        |
| + + +         | $60 \cdot 10^6$ | 10           | 1                        |
| Center        | $40 \cdot 10^6$ | 5            | 0.1                      |
| Outlier       | $40 \cdot 10^6$ | 20           | 2                        |
| Edge          | $40 \cdot 10^6$ | 0            | 0.01                     |

concentration were determined using a blood analyser (SuperGL compact, Hitado GmbH, Germany). Product concentration was measured on an Agilent 1200 HPLC system (Agilent Technologies, Santa Clara, CA) using a Protein A affinity column model 2-1001-00 (Applied Biosystems, Foster City, CA). A phosphate buffer (10 mmol/L  $NaH_2PO_4$ , 150 mmol/L NaCl (pH 7.5) were used as loading buffer and 10 mmol/L HCl, 150 mmol/L NaCl (pH 2.0) was used as elution buffer, respectively. Ammonia concentrations were evaluated using the Ammonia (Rapid) Assay kit (Megazyme, Ireland) according to the manufacturer's protocol. Nucleotides and nucleotide sugar concentrations were determined by two orthogonal techniques, high-pressure anion exchange chromatography (HPAEC), and MALDI-TOF MS, as described in Supplementary section S2.

### N-Linked Glycan Analysis

The monoclonal antibody N-linked glycosylation pattern was analyzed using capillary gel electrophoresis (CGE-LIF, DNA genetic analyzer 3130XL, Life Technologies, Germany). Antibody eluate from small-scale affinity columns (Phytips<sup>®</sup>, PhyNexus) was separated using capillary gel electrophoresis and detected with laser-induced fluorescence as described in Papac et al. (1998) and Rapp et al. (2011).

## Mathematical Model

### Mechanistic Model of N-Linked Glycosylation

#### Model Description

A mechanistic mathematical model was developed based on the work of Villiger et al. (2016b) and Jimenez del Val et al. (2011), and adapted for the simulation of the glycosylation profile of mAbs produced in steady state perfusion cell cultures (detailed description in Supplementary section S1). Briefly, the modeling framework can be divided in two sections. The first part comprises a simple unstructured cell culture model for the estimation of cell specific parameters (e.g., growth rate, ammonia and antibody specific productivities) and process related values (e.g., bleed rate). The mass balances of the perfusion bioreactor are based on the assumption of a CSTR reactor (e.g., perfect mixing). The equations

have been adapted to perfusion mode from Villiger et al. (2016b) and are summarized in Table II (Equations [1]–[10]). Variables derived from this set of equations provide initial and boundary conditions for the second section of the model, where the intracellular N-linked glycosylation process is simulated.

The glycosylation model is based on the cisternal maturation assumption with recycling of Golgi resident proteins first proposed by Jimenez del Val et al. (2011). The mass balances for the species involved were calculated considering the Golgi apparatus as a single plug flow reactor operating at steady state (structural parameters in Table SI) according to equations (11)–(13) in Table II and the reaction scheme reported in Figure 1 (Villiger et al., 2016b). The effect of the operating conditions on N-glycosylation was simulated by linking the kinetics of enzymatic reactions to the availability of sugar precursors, and the concentrations of metal ions and ammonia (Table SIII, equations [15]–[19]) (Villiger et al., 2016b). In detail, the impact of ammonia was related to the pH in the Golgi apparatus, which affects the enzyme activities. Galactose supplementation was linked to synthesis and incorporation of UDP-Gal, while dissociation constants for the enzyme-cofactor complexes were included for GnTI, GnTII (assumed to be the same), and GalT to reproduce the effect of different Mn concentrations. Additional information, including transport equations, Golgi enzyme distributions, and model constants are reported in Supplementary section S1. Specific production rates of antibody (qmAb) and ammonia (qAMM) as well as nucleotide and nucleotide sugar concentrations (except UDP-Gal) were taken as input variables rather than introducing a new set of equations to compute them. The addition of such equations would add unnecessary complexity without improving the descriptive and predictive capability of the model. A detailed analysis of this assumption is available in Supplementary section S4.

### Parameter Estimation

Parameter estimation was formulated as least squares minimization and performed in MATLAB R2014b (Mathworks, Natick, MA) with the MEtaheuristics for bioinformatics Global Optimization (MEIGO) toolbox (Egea et al., 2014), using the enhanced scatter search method and function *fmincon* as local MATLAB solver. The first eight runs of the DoE shown in Table I were used as a training dataset to estimate the model parameter values summarized in Table III. Parameter uncertainty was computed with the nonlinear regression algorithm provided by the *nlparci* function of MATLAB using the numerically calculated Jacobian matrix. Model output uncertainty was evaluated by running the model with 1000 sets of normally distributed parameter values. A local sensitivity analysis was performed computing the impact of differential changes in single estimated parameter values on the solution behavior.

### Response Surface Model

For comparison, the same set of experimental data was applied to generate an empirical linear regression model containing intercept, linear terms, and pairing terms for the different predictors investigated (Eq. 14).

**Table II.** Mass balance equations for the unstructured cell culture model and the mechanistic glycosylation model adapted from (Villiger et al., 2016b) for perfusion bioreactors. Enzyme kinetics  $r_j$  and nucleotide transport equations  $F_{T,k}$  are summarized in Tables SIII and SIV.

| Description                                                      | Equation                                                                                                                                                            | Number |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <i>Cell culture model</i>                                        |                                                                                                                                                                     |        |
| Volume                                                           | $\frac{dV}{dt} = F_{in} - F_{out}$                                                                                                                                  | (1)    |
| Bioreactor mass balance                                          | $F_{out} = F_{bleed} + F_{harvest}$                                                                                                                                 | (2)    |
| Bleed rate                                                       | $F_{bleed} = \mu V$                                                                                                                                                 | (3)    |
| Viable cell density                                              | $\frac{d(V[X_v])}{dt} = \mu V[X_v] - F_{bleed}[X_v]$                                                                                                                | (4)    |
| Growth rate                                                      | $\mu = \mu^{\max} \left( \frac{K_{\mu,amm}}{K_{\mu,amm} + [AMM]} \right)$                                                                                           | (5)    |
| Mn concentration                                                 | $\frac{dV[Mn]}{dt} = F_{in}[Mn_{in}] - F_{out}[Mn]$                                                                                                                 | (6)    |
| Ammonia concentration                                            | $\frac{d(V[AMM])}{dt} = q_{amm} V[X_v] - F_{out}[AMM]$                                                                                                              | (7)    |
| mAb concentration                                                | $\frac{dV[mAb]}{dt} = q_{mAb} V[X_v] - F_{out}[mAb]$                                                                                                                | (8)    |
| UDP-Gal balance with external addition of galactose <sup>a</sup> | $\frac{d[NS_k^{cyt}]}{dt} = m_{NS,k} + k_{NS,k} \left( [NS_k^{cyt}]_{sat} - [NS_k^{cyt}] \right)$                                                                   | (9)    |
| Galactose/UDP-Gal equilibrium <sup>a</sup>                       | $[NS_k^{cyt}]_{sat} = \frac{[MS_k]_{in}}{K_{NS,k}^{MS}}$                                                                                                            | (10)   |
| <i>Glycosylation model</i>                                       |                                                                                                                                                                     |        |
| Oligosaccharides mass balance                                    | $\frac{\partial [OS_i(z, t)]}{\partial t} = -\frac{4Q}{\pi D^2} \frac{\partial [OS_i(z, t)]}{\partial z} - \sum_{j=1}^{N.R.} \nu_{ij} r_j$                          | (11)   |
| Nucleotide sugars mass balance                                   | $\frac{\partial [NS_k^{golgi}(z, t)]}{\partial t} = F_{T,k} - \frac{4Q}{\pi D^2} \frac{\partial [NS_k^{golgi}(z, t)]}{\partial z} - \sum_{j=1}^{N.R.} \nu_{kj} r_j$ | (12)   |
| Nucleotides mass balance                                         | $\frac{\partial [N_k^{golgi}(z, t)]}{\partial t} = -F_{T,k} - \frac{4Q}{\pi D^2} \frac{\partial [N_k^{golgi}(z, t)]}{\partial z} + \sum_{j=1}^{N.R.} \nu_{kj} r_j$  | (13)   |

<sup>a</sup>With NS = UDPGal,  $k$  = Gal, and MS = Gal.

$$Y = \beta_0 + \sum_{i=1}^k \beta_i x_i + \sum_{i=1}^k \sum_{j=1}^{i-1} \beta_{ij} x_i x_j \quad (14)$$

with  $Y$  representing the fraction of individual glycan structures,  $x_i$  the variables investigated (i.e., VCD, manganese, and galactose), and  $\beta_i$  the corresponding adjustable coefficients. The RSM was generated in MATLAB using the built in “*stepwiselm*” function. Stepwise regression was used to eliminate coefficients whose 95% confidence intervals included the value zero (Draper and Smith, 1998).

## Results and Discussion

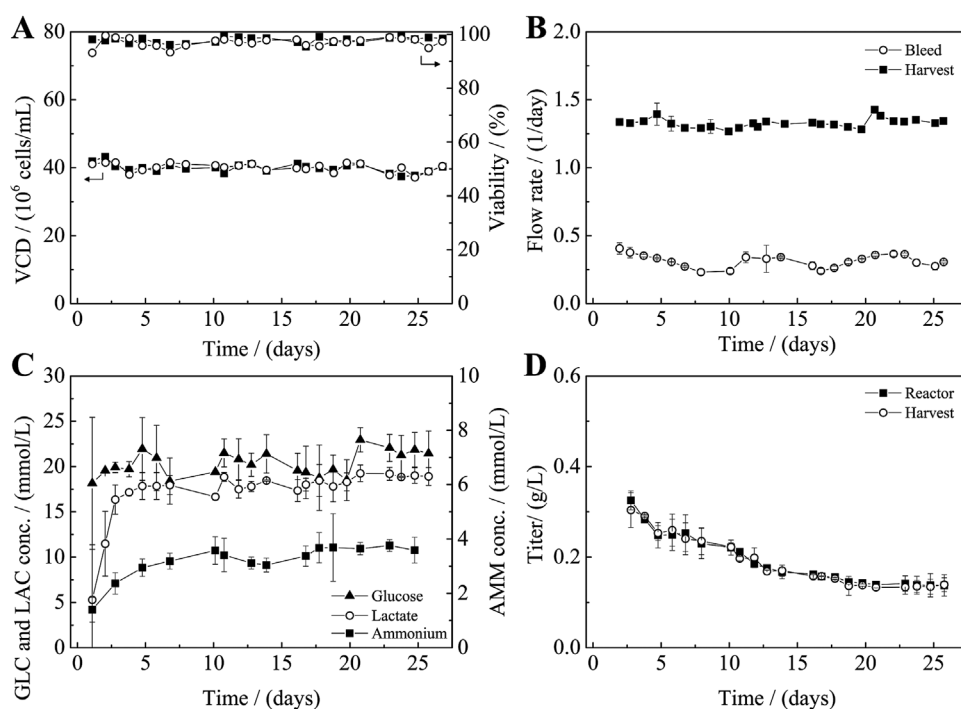
### Perfusion Culture

To investigate the long-term impact of the perfusion mode on the N-linked glycosylation of a monoclonal antibody, biological duplicates of a 26 days culture run at constant VCD were performed. The corresponding data on cell density, viability, main metabolites, and mAb concentration are shown in Figure 2. CHO cells were inoculated close to the set point VCD of  $40 \times 10^6$  cells/mL from a perfusion seed bioreactor to exclude the initial growth phase. The VCD set point was maintained with high viability in both cultures for the entire duration of the

**Table III.** Parameter estimation and confidence intervals for the unstructured perfusion cell culture and the N-glycosylation models.

| Description                                            | Variable                 | Unit                  | Value                 | 95%CI                                        |
|--------------------------------------------------------|--------------------------|-----------------------|-----------------------|----------------------------------------------|
| <i>Cell culture model</i>                              |                          |                       |                       |                                              |
| Maximum growth rate                                    | $\mu^{\max}$             | (1/d)                 | $9.09 \times 10^{-1}$ | $[4.55 \times 10^{-1}; 1.36]$                |
| Ammonia growth inhibition                              | $K_{\mu, \text{amm}}$    | (mmol/L)              | 1.86                  | $[1.31 \times 10^{-1}; 3.60]$                |
| Ammonia specific productivity                          | $q_{\text{amm}}$         | (mmol/cell/d)         | $9.36 \times 10^{-2}$ | $[6.43 \times 10^{-2}; 1.13 \times 10^{-1}]$ |
| Antibody specific productivity                         | $q_{\text{mab}}$         | (pmol/cell/d)         | 8.03                  | [7.20; 8.87]                                 |
| Maintenance coefficient of UDP-Gal                     | $m_{\text{UDP-Gal}}$     | (mmol/d)              | $6.16 \times 10^{-2}$ | $[5.71 \times 10^{-2}; 7.15 \times 10^{-2}]$ |
| Sugar mediated nucleotide sugar synthesis <sup>a</sup> | $k_{\text{UDP-Gal}}$     | (1/d)                 | $5.14 \times 10^{-1}$ |                                              |
| Equilibrium constant Gal/UDP-Gal                       | $K_{\text{Gal,UDP-Gal}}$ | (mmol/L)              | 5.69                  | $[4.80 \times 10^1; 71.5 \times 10^1]$       |
| <i>Glycosylation model</i>                             |                          |                       |                       |                                              |
| Dissociation constant for ManI (M7)                    | $K_{d, \text{ManI}, 7}$  | ( $\mu\text{mol/L}$ ) | $1.59 \times 10^3$    | $[1.58 \times 10^3; 1.59 \times 10^3]$       |
| Dissociation constant for GnT II                       | $K_{d, \text{GnTII}}$    | ( $\mu\text{mol/L}$ ) | $9.71 \times 10^2$    | $[9.64 \times 10^2; 9.78 \times 10^2]$       |
| Dissociation constant for GalT                         | $K_{d, \text{GalT}}$     | ( $\mu\text{mol/L}$ ) | $1.16 \times 10^4$    | $[1.14 \times 10^4; 1.17 \times 10^4]$       |
| Dissociation constant for SiaT                         | $K_{d, \text{SiaT}}$     | ( $\mu\text{mol/L}$ ) | $3.81 \times 10^4$    | $[3.75 \times 10^4; 3.87 \times 10^4]$       |
| Manganese dissociation constant for GnT                | $K_{\text{Mn, GnT}}$     | ( $\mu\text{mol/L}$ ) | $5.47 \times 10^{-3}$ | $[5.47 \times 10^{-3}; 5.47 \times 10^{-3}]$ |
| Manganese dissociation constant for GalT               | $K_{\text{Mn, GalT}}$    | ( $\mu\text{mol/L}$ ) | $3.82 \times 10^{-2}$ | $[3.78 \times 10^{-2}; 3.86 \times 10^{-2}]$ |
| Ammonia associated Golgi pH constant                   | $N_A^{\text{golgi}}$     | (mmol/L)              | $4.57 \times 10^1$    | $[4.57 \times 10^1; 4.57 \times 10^1]$       |

<sup>a</sup>Taken from (Villiger et al., 2016b).



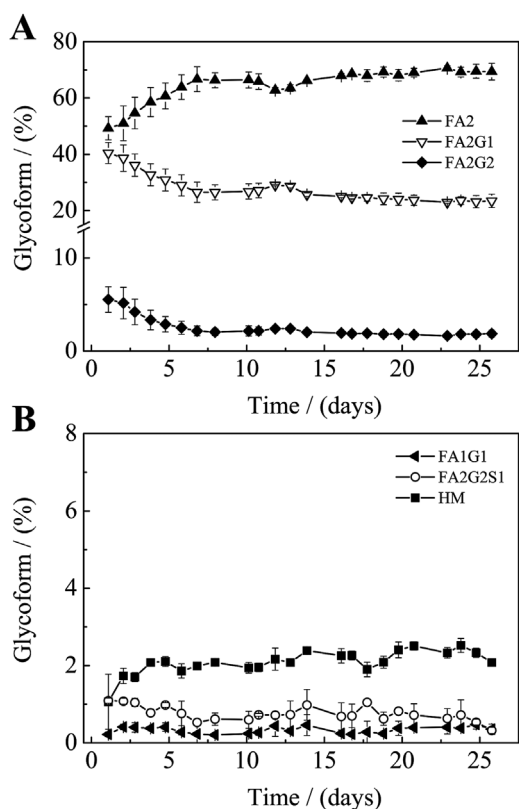
**Figure 2.** Culture data of duplicate perfusion culture runs at  $40 \times 10^6$  cells/mL. Time profiles of (A) cell density and viability for constant run 1 (■) and 2 (○), (B) average main flow rates (harvest and bleed), (C) main metabolite concentrations (glucose, lactate, and ammonia), and (D) antibody concentration in reactor and harvest ( $n = 2$ ).

experiments (Fig. 2A). The main flow rates of the perfusion are shown in Figure 2B. The harvest rate was fixed at 1.33 RV/day, while the bleed rate was adjusted automatically with the use of the online VCD measurement to exclude set point exceeding cells. Given its dependence on cellular growth (specific growth rate shown in Fig. S1A), the bleed flow rate remained at an average value of 0.33 RV/day with very small variations. Levels of the main metabolites glucose, lactate, and ammonia (Fig. 2C) remained stable after an initial transition phase, most possibly due to the inoculation of cells from the seed bioreactor. While only slight variation of glucose and lactate were observed after day 3, ammonia concentrations required up to 5/6 days to reach stable values. Eventually, glucose and lactate values leveled out around 20 mmol/L. Ammonia remained constant below 4 mmol/L throughout the culture, a non-toxic level for CHO cells (Yang and Butler, 2000). Overall metabolic consumption and production rates remained unchanged during steady operation (Fig. S1B). These data suggest, that the precise control of the perfusion bioreactor operating conditions, especially flow rates, VCD, and inlet media composition, enabled a constant cellular behavior, and resulted in a quasi-steady state of the culture. For the sake of simplicity, this state of bioreactor operation will be referred to as steady state in the following. The monoclonal antibody concentration in the harvest and reactor is shown in Figure 2D. Although no retention in the hollow fiber module was observed, the titer concentration declined during the first 10 days of the culture. The loss of protein expression with culture time, as indicated by the reduced specific mAb productivity (Fig. S1C), is a known problem during long-term cultures (Bailey et al., 2012; Ozturk

and Pálsson, 1990), and can be accounted to cell line instability or unfavorable media composition. It is worth noting, that the cell line under investigation has not been optimized for long term culture and no selection agent was used in the seed nor the production bioreactors.

### mAb Glycosylation at Steady-State Operation

The time evolution of N-glycan moieties measured by CGE-LIF during the steady state operation described above is shown in Figure 3. FA2 and FA2G1 represented the major oligosaccharide residues, whereas fully galactosylated, sialylated, and high mannose forms remained below 5%. Following the seed bioreactor, the changing cellular environment (Fig. 2C) resulted in a considerable initial variation of the N-linked glycosylation pattern. Given the exposure to the new media composition and increasing by-products levels, the distribution is shifted from more complex (FA2G1 and FA2G2) toward less processed forms (HM and FA2) with time. Ammonia has been frequently reported to reduce galactosyltransferase (GalT) and other glycosylation enzyme activities along the glycan processing machinery (Gawlitze et al., 2000, 2009) and can explain the observed behavior. Following the equilibration of metabolite concentrations (Fig. 2C), the N-glycosylation pattern evolved into a steady state. As described in Perfusion Culture section, the stable physiological environment of cells clearly established the consistent glycosylation of mAbs. Average values of HM, FA2, FA2G1, FA2G2, and FA2G2S1

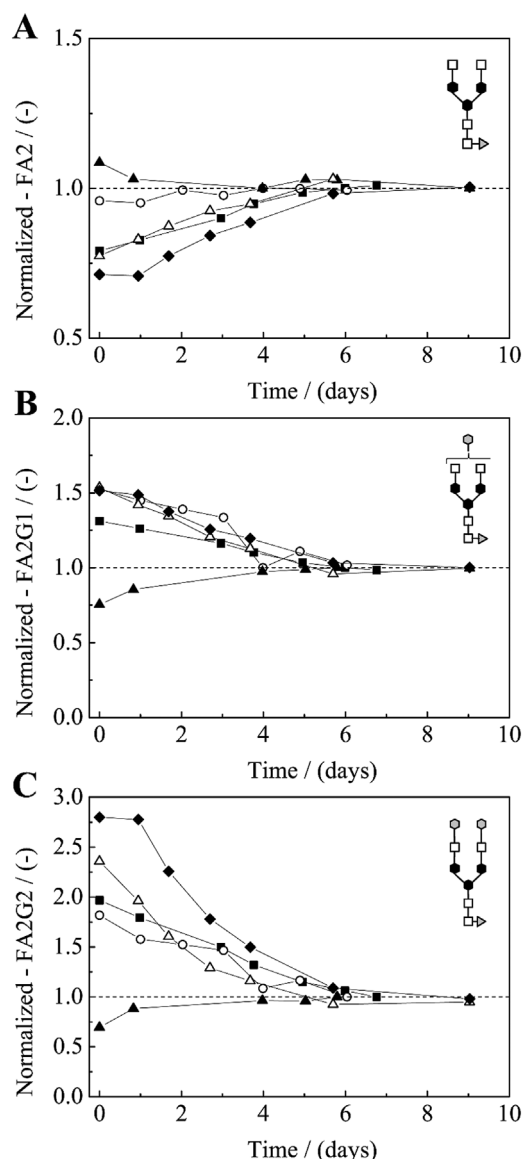


**Figure 3.** Evolution of N-linked glycan moieties split into main (A) and minor (B) species during the constant culture run at  $40 \times 10^6$  cells/mL ( $n=2$ ).

glycan after 6 days were  $2.2 (\pm 0.25)\%$ ,  $67.1 (\pm 2.8)\%$ ,  $25.6 (\pm 2.5)\%$ ,  $2.0 (\pm 0.4)$ , and  $0.68 (\pm 0.29)$ . No effect of the reduced specific protein production (Fig. S1C) was observed on glycosylation, although decreased protein secretion has been linked to higher glycoform complexity for some cell lines before (Fan et al., 2015; Jimenez del Val et al., 2016). Hence, the impact of the continuous operation mode and the metabolite composition in the cell environment was significantly higher compared to the observed change in specific productivity (Fig. S1C). Although VCD and flow rates have been monitored and controlled precisely, small perturbations in key process variables and cell intrinsic factors such as metabolism and gene expression, which ultimately impact glycosylation are possible. Despite these effects, the variability of the glycan distribution observed over time in the perfusion cultures does not exceed 3% for the three main forms FA2G0, FA2G1, and FA2G2. For comparison, during fed-batch cultures, these variations can exceed 20% for the main forms on a 10 days period (Villiger et al., 2016a). The reproducibility of the experimental data was assessed through the relative standard deviation (RSD) defined as the ratio of the glycan fraction standard deviation to its average fraction at steady state for the two constant cultures. The obtained RSD values of 2.5%, 6.2%, and 13.0% for FA2, FA2G1, and FA2G2, respectively indicate good reproducibility. It is worth noting that for glycan

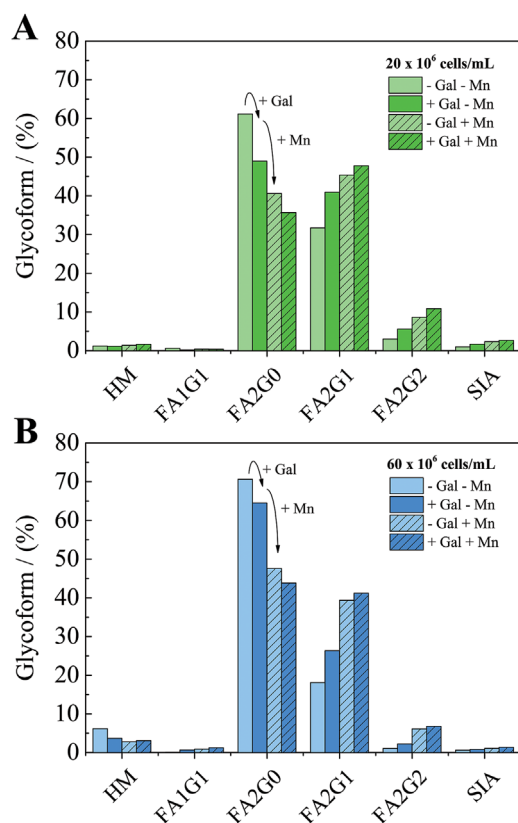
structures with small absolute abundance the intrinsic instrument accuracy becomes dominant.

To further validate this result, the time evolution of the main glycan fractions for the beginning of the  $40 \times 10^6$  cells/mL constant culture (Fig. 3) and three exemplary cultures of the DoE sequential screening approach (i.e., — — —, + — — and *center point* in Table I) are illustrated in Figure 4. After an initial incline or decline the fractions converged to a steady state value until working day 6 and remained constant thereafter. This time frame of 6 days from constant cell density has also been observed by other groups (Xu and Chen, 2016) and can be attributed to extrinsic factors such as the medium exchange kinetics of the stirred tank reactor as



**Figure 4.** Time evolution of glycan forms FA2 (A), FA2G1 (B), and FA2G2 (C) for several experimental conditions (■ DoE (— — —), ○ DoE (+ — —), ▲ DoE (center point), Δ constant run 1, ◆ constant run (2) normalized to their steady state value. Time starts when the VCD setpoint has been achieved.





**Figure 5.** Steady state values of DoE conditions as affected by galactose and manganese supplementation, grouped for VCD of (A)  $20$  and (B)  $60 \times 10^6$  cells/mL. Plus (+) and minus (–) signs indicate high and low levels of corresponding supplements as reported in Table 1.

well as cell-intrinsic mechanisms (changes in gene expression and metabolism).

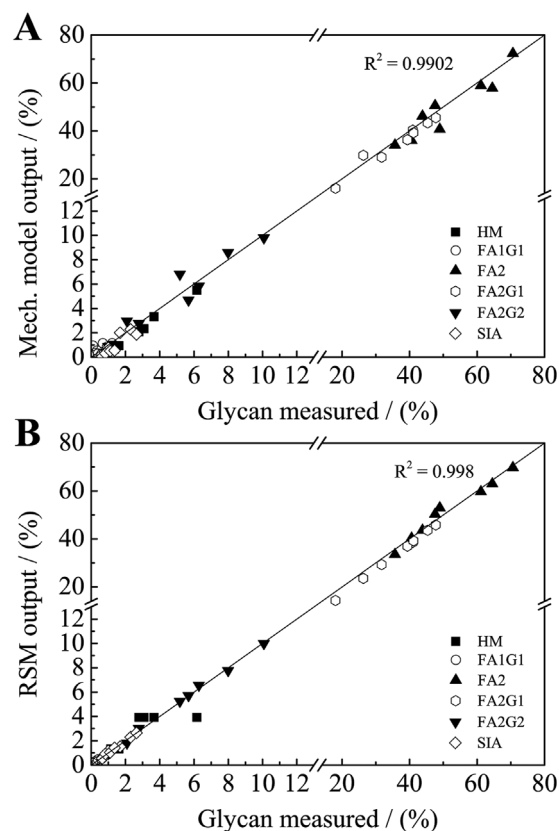
### Modulation of N-Linked Glycosylation in Perfusion Experiments

As mentioned above, a body-centered DoE was applied to investigate the effect of manganese, galactose, and VCD on N-linked glycosylation of mAbs. Multiple culture experiments employing identical bioreactor setups were carried out to screen the operating conditions of the DoE in a sequential manner. Each condition was kept constant for at least 7 days to allow the convergence of the cell culture to its quasi-steady state value. The sequential screening approach allowed testing of multiple operating points in a single culture, minimizing the time, media, and labor involved, while maximizing experimental reproducibility. Typically, three to four set points were included in one culture experiment. An exemplary VCD profile of varying set points is shown in Figure S2A. Bioreactors were operated according to a previous physical characterization (Karst et al., 2016), which resulted in suitable process parameters in terms of gas-liquid mass transfer and hydrodynamics to support high cell density. In all culture experiments, the perfusion rate was given by the constant harvest rate of 1 RV/day and the adaption of the bleed rate according to an online biomass sensor (Fig. S3). Stable cell density and high viability ( $>90\%$ ) was

achieved independent of the operating point. A more detailed discussion of the cell behavior at these different operation points is given in reference (Karst et al., 2016). Depending on the VCD set point a change in ammonia levels between  $1.5 \pm 0.12$  and  $7.1 \pm 1.6$  mmol/L at  $20$  and  $60 \times 10^6$  cells/mL was observed, respectively (Fig. S2B). Galactose (low:  $0$  and high:  $10$  mmol/L) and  $\text{MnCl}_2$  (low:  $0.01$  and high:  $1 \mu\text{mol/L}$ ) were supplemented to the fresh media feed accordingly and were shown not to affect the growth kinetics of cells (data not shown).

The N-glycosylation pattern for each condition, illustrated in Figure 5 (values listed in Table SXI), were sampled on the final days of each DoE point to ensure that glycosylation converged to its steady state value. A clear trend was detected for all evaluated conditions. As expected, a significant increase in mAb galactosylation was observed, when cell media were supplemented with galactose or manganese. A switch from  $0.01$  (Low) to  $1 \mu\text{mol/L}$  (High) Mn in presence of low ammonia concentrations at  $20 \times 10^6$  cells/mL resulted in a  $14\%$  and  $6\%$  increase of the relative concentration in FA2G1 and FA2G2 forms, respectively, while the increase in the presence of  $0$  (Low) to  $10$  mM (High) galactose was  $9\%$  and  $2\%$ . Similar trends were observed at high ammonia levels at  $60 \times 10^6$  cells/mL. A synergic effect was detected when the two substances were added simultaneously.

High mannose structures were always found to increase at high VCD due to higher ammonia concentrations, but overall remained well below  $10\%$ . Similarly, sialylated glycans were present in higher



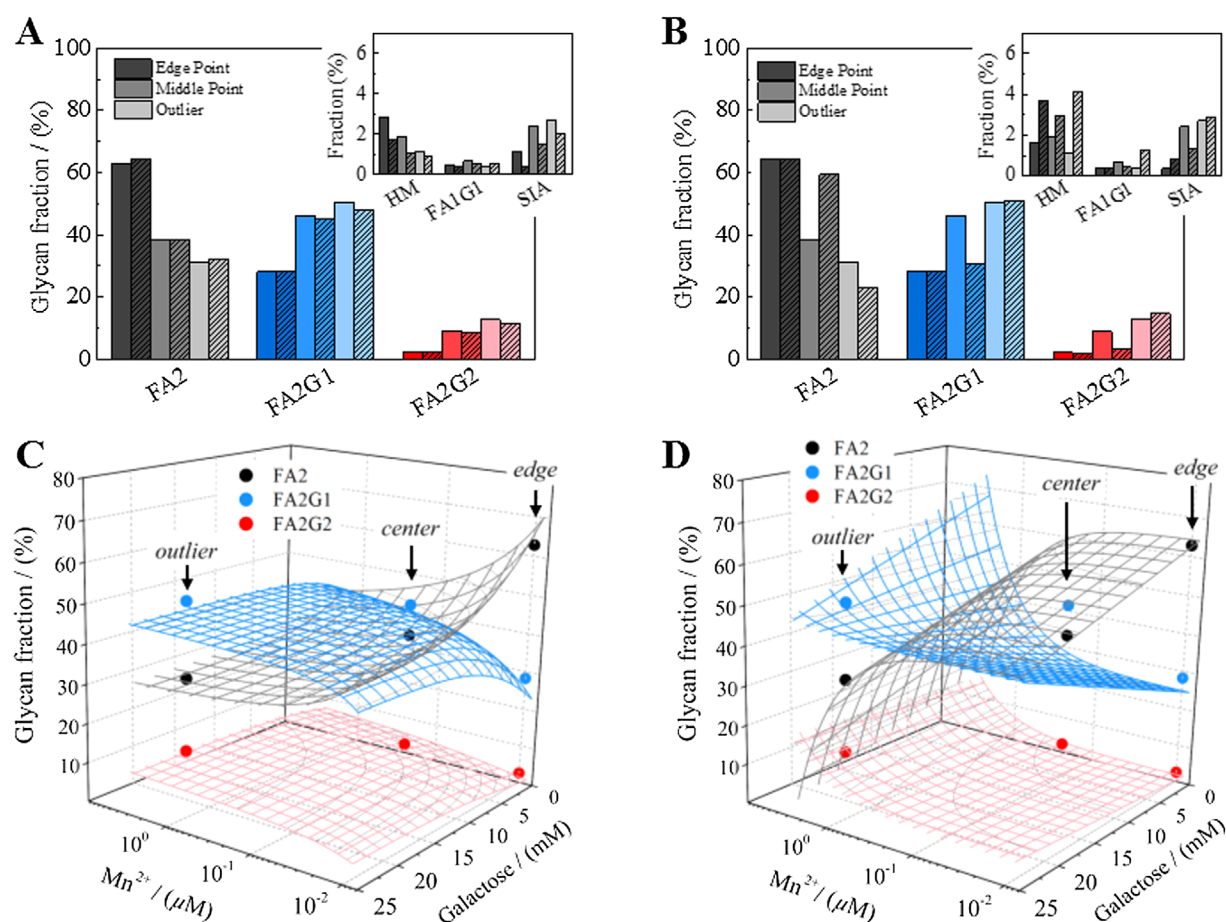
**Figure 6.** Comparison of experimentally measured glycan fractions to the model output of the mechanistic N-glycosylation model (A) and the empirical RSM model (B).

amounts at  $20 \times 10^6$  cells/mL, although they never exceeded 2% of the total glycosylation profiles. The higher ammonia concentrations observed at  $60 \times 10^6$  cells/mL led to generally less processed forms (FA2G1, FA2G2, and sialylated glycans), when compared to  $20 \times 10^6$  cells/mL set points. The manipulation of VCD, manganese, and galactose supplementation allowed a potential for modulation between the minimum ( $60 \times 10^6$  cells/mL,  $0.1 \mu\text{M}$  Mn,  $0 \text{ mmol/L}$  Gal) and maximum ( $20 \times 10^6$  cells/mL,  $1.0 \mu\text{mol/L}$  Mn,  $10 \text{ mmol/L}$  Gal) from 70% to 36% in FA2, 18% to 48% in FA2G1 and 1.1% to 11% in FA2G2, respectively. Depending on the combination of the three parameters a desired glycosylation pattern is achievable and can be kept within low variability, when operating the culture in steady-state. To support such a systematic modulation approach, a mechanistic model was trained with the underlying dataset and used to predict new experimental conditions and corresponding glycosylation patterns.

### Modeling of N-Linked Glycosylation

The performance of both the mechanistic model and the RSM were assessed by comparing the steady state glycosylation profiles

obtained experimentally with the results of the simulations. The tuning of the model parameters for the two strategies was performed on the first eight DoE experiments listed in Table I. As shown in Figure 6A, the mechanistic model simulations show good agreement with the experimental glycan structures measured in the DoE experiments even for highly modulated structures such as FA2 and FA2G1. These data suggest that the model is capable to reproduce both the inhibitory effect of ammonia as well as the effects of Mn and galactose in perfused conditions, according to the overall trends discussed in the previous sections. The fitting parameters of the mechanistic model and their 95% confidence intervals are summarized in Table III. Six cell-specific parameters provided the correct cellular environment, in terms of growth ( $\mu^{\text{max}}, K_{\mu, \text{AMM}}$ ), galactose synthesis ( $m_{\text{UDP-Gal}}$ ) and incorporation ( $K_{\text{Gal, UDP-Gal}}$ ), ammonia ( $q^{\text{AMM}}$ ), and mAb specific productivity ( $q^{\text{mAb}}$ ), which constituted the input conditions for the glycosylation model. For the glycosylation model, the estimation of seven parameters was needed to correctly reproduce the glycosylation profiles observed in the entire DoE. Six parameters are dissociation constants for Golgi resident enzymes, while the term  $N_A^{\text{Golgi}}$  was used to relate the transport of ammonia to the Golgi apparatus according to Table SIII,



**Figure 7.** Comparison of the mechanistic (A) and empirical RSM (B) model predictions to the experimentally determined glycan distribution at the center (dark), edge point (medium), and outlier (light). Plain bars represent experimental data and striped bars represent the corresponding model predictions. Surface representation of the model prediction at  $40 \times 10^6$  cells/mL as function of manganese and galactose supplementation for single glycan forms FA2G0 (black), FA2G1 (blue), and FA2G2 (red) for the mechanistic (C), and empirical RSM (D) model. Points indicate experimental values of center, edge point, and outlier.

Equations (4)–(5). These parameters were used to fit a total of 48 glycosylation structures with an average deviation error of less than 9% per glycosylation profile, in line with other models published in literature (Hossler et al., 2007; Jimenez del Val et al., 2011; Krambeck and Betenbaugh, 2005). All parameters were found to be practically identifiable and had significant impact on different structures (Fig. S8).

As shown in Figure 6B, the stepwise fitting of the RSM also provided a good description of the experimental data. Depending on the glycan structure, the polynomial equations consisted of combinations of significantly contributing variables and cross terms (Table SXII). For example, HM structures could be described only with the linear term as function of the VCD, whereas FA2G1 required two interaction terms in addition to all three linear terms (Mn, Gal, and VCD). The presence of linear terms for the manipulated variables in the equations of all other structures, confirmed that these had a significant impact on glycosylation.

### Model Reliability and Predictive Capability

The model's predictive capability was validated with an independent set of experimental conditions located both inside and outside the experimental space used for parameter estimation. The conditions investigated for this aim include  $40 \times 10^6$  cells/mL without additional Mn and Gal supplementation (*edge point*),  $40 \times 10^6$  cells/mL supplemented with 5 mmol/L galactose and 0.1  $\mu$ mol/L Mn (*center point*), and  $40 \times 10^6$  cells/mL supplemented with 5 mmol/L galactose and 0.1  $\mu$ mol/L Mn (*outlier*), as indicated in the last three rows of Table I. The obtained glycosylation profiles are consistent with the results of the previous runs inside the DoE, with increasing levels of Gal and Mn correlating with higher levels of terminal galactosylation and sialylation. In particular, the outlier presents over 50% of FA2G1 glycans and only 30% of FA2. As shown in Figure 7A, the experimental results were successfully predicted by the model (with the same set of parameter values), with small deviations only in case of low abundant structures as discussed previously. These results demonstrate the reliability of the model in both intra- and extrapolation from the operating conditions covered by the DoE. For comparison, the same experimental results were compared with the predictions of the statistical RSM (Table SXII again with no change in the parameter values). As shown in Figure 7B (white bars), the RSM provided a correct prediction only for the glycosylation profile of the *edge point* condition. Large deviations from the experimental results (up to 20% in case of FA2 and 16% for FA2G1) were registered while trying to interpolate the glycosylation for the *center point*. On the other hand, the prediction of the *outlier* surprisingly produced a better result, although the errors in FA2 and HM glycoform predictions are very significant (20% for the first one, 2.5% for the latter). This is confirmed by the prediction surfaces for main glycan forms as a function of manganese and galactose at constant VCD of  $40 \times 10^6$  cells/mL shown in Figure 7C and D. When considering conditions outside the DoE parameter space, the mechanistic model predicts a logic saturation of main glycan structure fractions upon continuing supplementation of galactose and manganese. In contrast, the predictions of the RSM cannot be justified on a physical ground, as for high  $Mn^{2+}$  and Gal levels negative FA2 and unfeasible FA2G1

(>100%) fractions are predicted. This behavior is fully expected due to the intrinsic higher extrapolation capability of mechanistic over empirical models. In particular, since the mechanistic model is able to deliver valid predictions even after a limited amount of experiments. This allows us to conclude that mechanistic models can be reliably used for process development and optimization of therapeutic protein production in terms of product quality.

### Conclusion

Compared to traditional fed-batch operation, the stable cell environmental conditions of cells combined with low by-product levels and residence times are unique properties of continuous perfusion bioreactors. This justifies the renewed interest in this operational mode which is currently emerging for the production of therapeutic proteins. We have shown that the constant operation of flow rates, media composition, and VCD resulted in the consistent glycosylation of the expressed mAb for a time period of over 20 days. Independent of the loss in specific mAb productivity, the time resolved variability of the glycan distribution was significantly decreased compared to fed-batch operation. The continuous harvest of antibodies with desired and consistent N-linked glycosylation pattern is beneficial in sight of the recent regulatory initiatives restricting the degree of variability in critical quality attributes (CQA). Additionally, perfusion cultures represent a promising platform for the modulation of glycosylation, as the effect of the manipulating variables can be decoupled from the impact of accumulation or depletion of other substances, both in the intra- and extracellular *milieu*. According to a DoE investigation carried out in a sequential screening of operating conditions, the supplementation with either manganese or galactose resulted in higher galactosylation of the antibody product, with a synergic effect, when both are present. The impact of viable cell density on the glycan distribution can be mostly attributed to the increase of ammonia for higher VCD, generally resulting in less processed forms.

With the aim of developing a strategy to explore the glycosylation in response to process conditions while reducing the experimental effort, we compared the performance of a mechanistic model and an empirical RSM. Both models are able to satisfyingly fit the experimental data within the parameter space covered by the DoE, including the distribution of the main glycan structures. However, only the mechanistic model provided reliable results also outside the experimental space when used in the predictive extrapolation mode. On the other hand, the response surface showed large errors in the prediction of the DoE middle point due to the non-linearity of the system. We therefore conclude that perfusion reactors allow the continuous harvest of a desired N-linked glycosylation pattern within small variability, reducing the product quality heterogeneity, as well as the modulation of such a pattern by properly acting on the operating conditions and media composition.

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## Nomenclature

### Latin Letters

|                                  |                                                            |
|----------------------------------|------------------------------------------------------------|
| AMM                              | ammonium                                                   |
| $\beta_1$                        | response surface model coefficients                        |
| CGE-LIF                          | capillary gel electrophoresis—laser induced fluorescence   |
| $E_j$                            | enzyme $j$                                                 |
| $E_j^{\max}(z)$                  | peak concentration of $E_j$ at Golgi length $z$            |
| $F_{\text{in}}$                  | flow rate in to the reactor (feed)                         |
| $F_{\text{out}}$                 | total flow rate out of the reactor                         |
| $F_{\text{bleed}}$               | bleed rate                                                 |
| $F_{\text{harvest}}$             | harvest rate                                               |
| $F_{T,k}$                        | flowrate of $NS_k$ into the Golgi                          |
| Fc                               | fragment crystallizable (region)                           |
| GalT                             | $\beta$ -1,4 Galactosyltransferase                         |
| GnT I                            | $\alpha$ -1,3 N-acetylglucosaminyl transferase I           |
| GnT II                           | $\alpha$ -1,6 N-acetylglucosaminyl transferase II          |
| HM                               | high mannose glycan moieties                               |
| IgG                              | immunoglobulin G                                           |
| $k_{f,j}^{\max}$                 | maximum turnover rate for reaction $j$                     |
| $K_{\mu,amm}$                    | ammonia growth inhibition constant                         |
| $K_{d,k}$                        | dissociation constant of the donor-enzyme complex          |
| $K_{d,Mn}$                       | dissociation constant of the manganese-enzyme complex      |
| $K_{d,Nk}$                       | dissociation constant of the $N_k$ -enzyme complex         |
| $K_{d,z}$                        | dissociation constant of the competitor-enzyme complex     |
| $k_{T,k}$                        | transport turnover rate                                    |
| $k_{NS,k}$                       | sugar mediated nucleotide sugar synthesis                  |
| $m$                              | maintenance coefficient                                    |
| $Mn$                             | manganese chloride                                         |
| $MS_k$                           | monosaccharide concentration in the medium                 |
| mAb                              | monoclonal antibody                                        |
| MALDI-TOF                        | matrix assisted laser desorption/ionization time-of-flight |
| Man I                            | $\alpha$ -1,2 mannosidase I                                |
| Man II                           | $\alpha$ -1,3/ $\alpha$ -1,6 mannosidase II                |
| MS                               | mass spectrometry                                          |
| $N_A^{\text{golgi}}$             | ammonia associated Golgi pH constant                       |
| $N_k$                            | nucleotide $k$                                             |
| $NS_k$                           | nucleotide sugar $k$                                       |
| $OS_i$                           | oligosaccharide structure $i$                              |
| $OS_z$                           | competing oligosaccharide substrates                       |
| $pH_{\text{opt}}$                | optimal culture pH                                         |
| $pH_{\text{opt}}^{\text{golgi}}$ | optimal Golgi pH                                           |
| $pH_{\text{opt}}^{\text{golgi}}$ | golgi pH                                                   |
| $pH_A^{\text{golgi}}$            | apparent $pK_A$ value of the Golgi                         |
| $Q$                              | volumetric flow rate into the Golgi cisterna               |
| $q_{mAb}^{\max}$                 | maximum specific mAb productivity                          |
| $q_{mAb}$                        | specific mAb productivity                                  |
| $q_{AMM}$                        | specific ammonia production                                |
| $r_j$                            | rate of reaction $j$                                       |
| SiaT                             | $\alpha$ -1,6 Sialyltransferase                            |
| SIA                              | sialyated glycan moieties                                  |
| $TP_k$                           | transport protein $k$                                      |

|                  |                                                       |
|------------------|-------------------------------------------------------|
| $TP_k^{\max}(z)$ | peak concentration of $TP_k$ at Golgi length $z$      |
| UDP-Gal          | uridine diphosphate galactose                         |
| $V$              | volume                                                |
| VCD              | viable cell density                                   |
| WD               | working day                                           |
| $Y$              | response surface model glycan structure fractions [%] |
| $x_i$            | response surface model variables (VCD, Mn and Gal)    |
| $z_j^{\max}$     | localization of peak concentration of $E_j$           |
| $z_k^{\max}$     | localization of peak concentration $TP_k$             |

### Greek Letters

|                |                                                      |
|----------------|------------------------------------------------------|
| $\nu_{i,j}$    | stoichiometric coefficient of $OS_i$ in reaction $j$ |
| $\nu_{k,j}$    | stoichiometric coefficient of $NS_k$ in reaction $k$ |
| $\mu^{\max}$   | maximum growth rate                                  |
| $\theta$       | parameter                                            |
| $\omega_k$     | width of the concentration profile of $TP_k$         |
| $\omega_{f,j}$ | width of the pH activity profile of $E_j$            |
| $\omega_{mAb}$ | width of specific productivity of $E_j$              |
| $\omega_j$     | width of concentration profile of $E_j$              |

### Glycosylation Nomenclature

|         |                                                                                                |
|---------|------------------------------------------------------------------------------------------------|
| M9      | Man <sub>9</sub> GlcNAc <sub>2</sub>                                                           |
| M8      | Man <sub>8</sub> GlcNAc <sub>2</sub>                                                           |
| M7      | Man <sub>7</sub> GlcNAc <sub>2</sub>                                                           |
| M6      | Man <sub>6</sub> GlcNAc <sub>2</sub>                                                           |
| M5      | Man <sub>5</sub> GlcNAc <sub>2</sub>                                                           |
| A2      | GlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub>                                       |
| FA2     | GlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc                                   |
| FA2G1   | GalGlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc                                |
| FA26G1  | $\alpha(1-6)$ GalGlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc                  |
| FA23G1  | $\alpha(1-3)$ GalGlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc                  |
| FA2G2   | Gal <sub>2</sub> GlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc                  |
| FA2G2S1 | SiaGal <sub>2</sub> GlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc               |
| FA2G2S2 | Sia <sub>2</sub> Gal <sub>2</sub> GlcNAc <sub>2</sub> Man <sub>3</sub> GlcNAc <sub>2</sub> Fuc |

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